CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

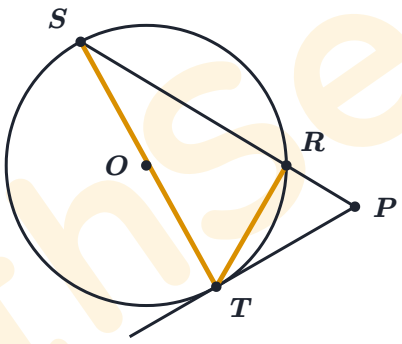
Chapter 18: Tangents and Intersecting Chords

Previous year board questions, topic-wise: 12 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

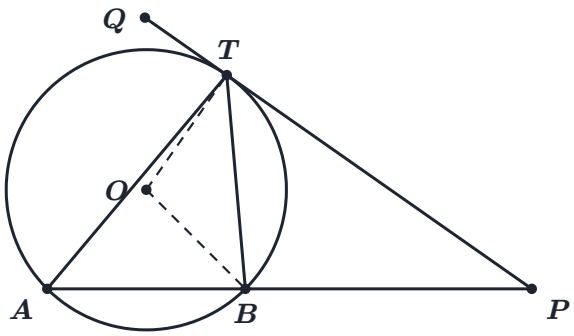
1 Tangent and secant from an external point

1. PT is a tangent to a circle at T, and a line through P cuts the circle at R and S, with R nearer to P. (a) Prove that $\triangle PTR \sim \triangle PST$. (b) Prove that $PT^2 = PR \times PS$. (c) If $PR = 4$ cm and $PS = 16$ cm, find PT. ICSE 2025, 3 marks



2. A circle has centre O and radius 9 cm. The diameter AB is produced to a point P, and PQ is a tangent from P. If $PA = 24$ cm, find PQ. ICSE 2024, 3 marks

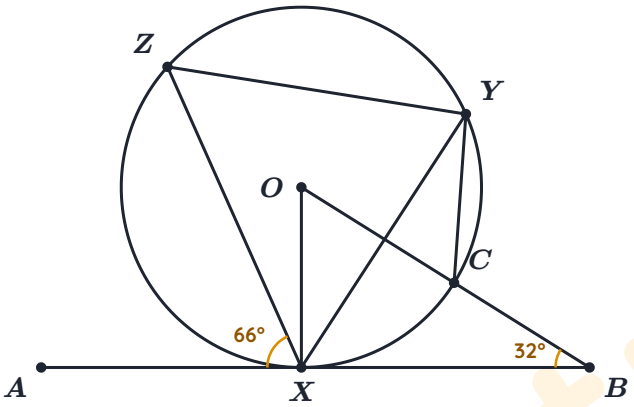
3. PQ is a tangent to a circle with centre O at T. The chord AB produced meets the tangent at P. $AB = 9$ cm, $BP = 16$ cm, $\angle PTB = 50^\circ$ and $\angle OBA = 45^\circ$. Find (a) PT (b) $\angle BAT$ (c) $\angle BOT$ (d) $\angle ABT$. ICSE 2023, 4 marks



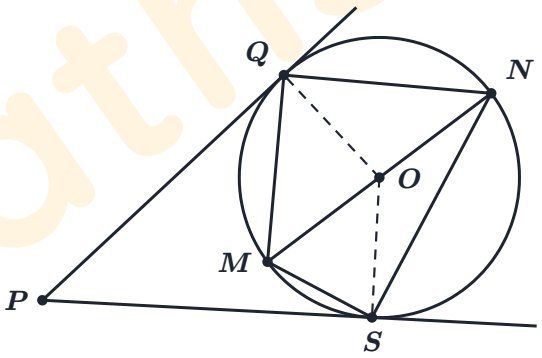
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2 The alternate segment theorem

4. X, Y, Z and C lie on a circle with centre O. AB is the tangent at X, OB cuts the circle at C, and $ZY = XY$. Given $\angle OBX = 32°$ and $\angle AXZ = 66°$, find (a) $\angle BOX$ (b) $\angle CYX$ (c) $\angle ZYX$ (d) $\angle OXY$. ICSE 2025, 4 marks

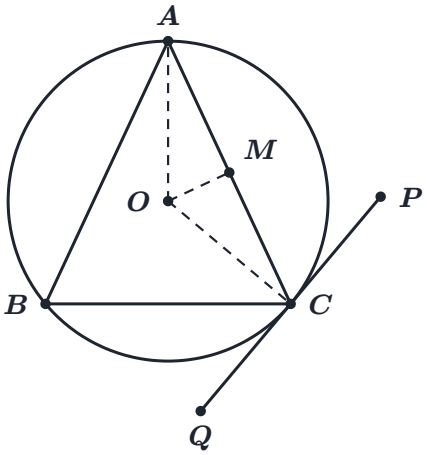


5. PR and PT are tangents from P to a circle with centre O, touching it at Q and S. MN is a diameter, with M on the arc QS nearer to P. Given $\angle PQM = 42°$ and $\angle PSM = 25°$, find (a) $\angle OQM$ (b) $\angle QNS$ (c) $\angle QOS$ (d) $\angle QMS$. ICSE 2024, 4 marks



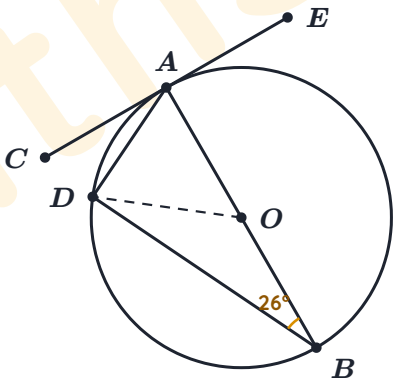
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6. An isosceles triangle ABC with $AB = AC$ is inscribed in a circle with centre O. PQ is the tangent at C, with Q on the side of BC away from A. OM is perpendicular to the chord AC and $\angle COM = 65°$. Find (a) $\angle ABC$ (b) $\angle BAC$ (c) $\angle BCQ$. ICSE 2024, 3 marks

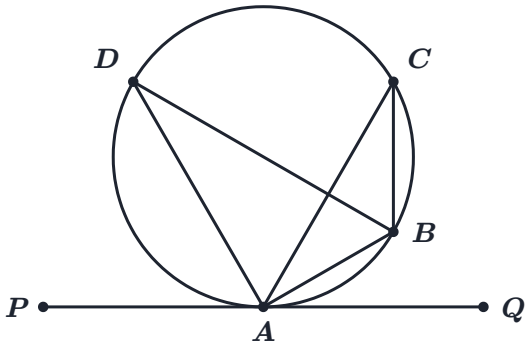


7. RT is a tangent to a circle at S. P and Q are points on the circle with $\angle PST = 30°$ and $\angle SPQ = 60°$. Find $\angle PSQ$. (a) $40°$ (b) $30°$ (c) $60°$ (d) $90°$ ICSE 2023, 1 mark

8. AB is a diameter of a circle with centre O, and CE is the tangent at A. D is a point on the circle on the same side of AB as C, with $\angle ABD = 26°$. Find (a) $\angle BDA$ (b) $\angle BAD$ (c) $\angle CAD$ (d) $\angle ODB$.
ICSE 2023, 4 marks



9. PQ is a tangent to a circle at A. B, C and D are points on the circle such that AB bisects $\angle CAQ$ and AD bisects $\angle PAC$. If $\angle BAQ = 30°$, prove that (a) BD is a diameter (b) $\triangle ABC$ is isosceles.
ICSE 2017, 3 marks

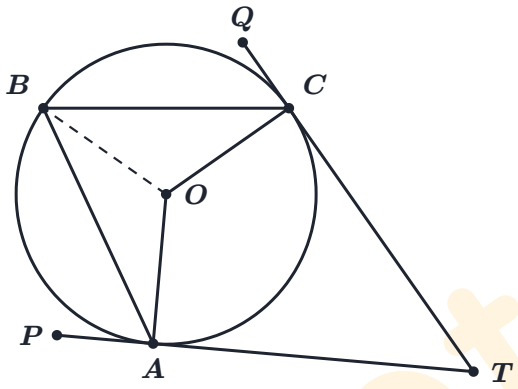


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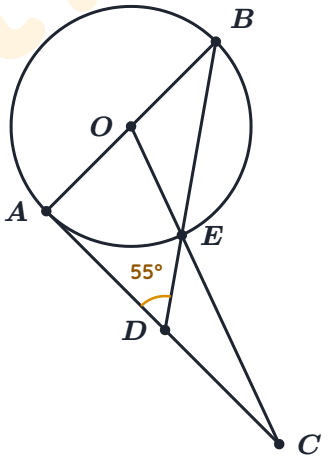
3 Tangent properties

10. PS and PT are tangents to a circle from P, with $\angle SPT = 80°$. Q is a point on the circle with $SQ \parallel PT$. Find $\angle QST$. (a) $140°$ (b) $90°$ (c) $80°$ (d) $50°$ ICSE 2024, 1 mark

11. TP and TQ are tangents to a circle with centre O, touching it at A and C. B is a point on the circle with $\angle BCQ = 55°$ and $\angle BAP = 60°$. Find (a) $\angle OBA$ and $\angle OBC$ (b) $\angle AOC$ (c) $\angle ATC$.
ICSE 2020, 4 marks



12. AB is a diameter of a circle with centre O, and AC is the tangent at A. D is a point on AC, and BD cuts the circle at E. The line OE produced meets the tangent at C. If $\angle ADB = 55°$, find $x = \angle ACO$ and $y = \angle AOE$. ICSE 2019, 3 marks



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